

Magnetometer Survey Protocol

Transects A-D / GPS-tagged field-strength baseline
Working Draft 0.1 / July 28, 2026

FIELD DRAFT: Final endpoints, land access, exclusion zones, and safety controls require field verification before collection. Transect C dimensions reflect the current survey specification.

1. PURPOSE

Create a repeatable, auditable baseline of total magnetic field readings across the Old Glory Peak corridor. The protocol links every reading to position, orientation, device state, geology, a sample identifier when collected, and a photographic record.

2. SURVEY DESIGN

Line	Target	Current specification	Primary purpose
A	Pinto Fault Crossing	Route perpendicular to mapped trace; final endpoints and spacing pending access reconnaissance.	Characterize field change across the fault corridor.
B	Morongo Valley Fault Crossing	Route perpendicular to mapped trace; final endpoints and spacing pending access reconnaissance.	Compare magnetic and water-context observations.
C	Old Glory Peak Ridge Crest	East-west; 400 m; 9 stations at 50 m spacing.	High-resolution mineralized-zone baseline near historical workings.
D	Mine Area Grid	Grid near historical adits; boundaries pending access and safety review.	Map local anomalies without entering workings.

3. REQUIRED EQUIPMENT

- Primary smartphone with three-axis magnetometer and raw-data export capability.
- Independent GNSS or GPS receiver when available; paper map and compass as backup.
- Non-magnetic measuring tape, station stakes or removable markers, field notebook, and time source.
- Camera; sample bags and labels; streak plate; Mohs kit; dilute HCl kit with eye and skin protection.
- Water test equipment when a lawful sample point is included: calibrated pH and ORP meters plus approved sample containers.
- First-aid kit, satellite messenger or radio, sun protection, water, and site-specific personal protective equipment.

4. DEVICE CONTROL AND CALIBRATION

- Record device model, operating-system version, application version, sensor range, and export format.
- Remove magnetic cases, watches, tools, vehicles, speakers, batteries, and other ferrous objects from the measurement area.
- At the daily control station, collect a 60-second stationary record before and after the transect.
- Run the device calibration motion only before the control record, never between ordinary stations unless the application reports loss of calibration.
- Maintain one device orientation throughout the survey; record any unavoidable orientation change as a protocol deviation.
- Synchronize timestamps to UTC and compare the control station with available USGS or geomagnetic reference data during quality review.

5. STATION SEQUENCE

Step	Field action	Record
1	Approach from the same direction and stop at the planned coordinate.	Station ID, UTC time, GPS accuracy.
2	Set device at the defined height and orientation; remain still for 30 seconds.	Height, azimuth, pitch, roll.
3	Capture three 10-second readings separated by 10 seconds.	Bx, By, Bz, total field, sample rate.
4	Repeat if any reading differs from the station median by more than the pre-registered tolerance.	Repeat flag and reason.
5	Describe lithology, visible structures, disturbance, weather, and nearby interference.	Field notes and controlled terms.
6	Take overview, ground, and close-detail photographs with a scale.	Photo IDs.
7	Collect a rock or water sample only when authorized and necessary.	Sample ID and chain of custody.

6. QUALITY CONTROL

- Use preassigned station IDs: OGP-[TRANSECT]-[SEQUENCE]. Never reuse an ID.
- Flag readings near vehicles, fencing, buried utilities, power lines, phones, tools, or other interference.
- Collect at least one duplicate station per ten ordinary stations and repeat the daily control station.
- Retain raw sensor files unchanged. Create cleaned and analysis-ready files as new versioned derivatives.
- Document every exclusion, correction, interpolation, or replacement in a machine-readable change log.
- Publish the protocol, field dictionary, and negative or null results with the data release.

7. CORE DATA DICTIONARY

Field	Format	Definition
station_id	text	Unique OGP transect and sequence identifier.
timestamp_utc	ISO 8601	Coordinated Universal Time at start of reading.
latitude / longitude	decimal degrees	WGS84 coordinate with horizontal-accuracy estimate.
bx / by / bz / total_nt	numeric	Magnetic components and total field in nanotesla.
orientation	degrees	Azimuth, pitch, and roll during collection.
lithology / mineral_id	controlled text	Field identification with confidence and later verification status.
photo_ids / sample_ids	text list	Linked photographs and chain-of-custody identifiers.
qc_flags	text list	Interference, repeat, calibration, or protocol-deviation flags.

8. FIELD SAFETY AND ETHICS

- Obtain landowner or agency authorization before entry, marking, sampling, or equipment placement.
- Do not enter adits, shafts, unstable mine workings, private enclosures, or posted exclusion zones.
- Use a two-person field team where terrain or communication conditions warrant it; maintain a check-in plan.
- Stop for lightning, extreme heat, wildfire conditions, hazardous wildlife, unstable footing, or loss of communication.
- Minimize disturbance, pack out all materials, and avoid publishing sensitive cultural or ecological coordinates.

9. RELEASE PACKAGE

Release the signed protocol, device inventory, station plan, raw export, cleaned table, QC log, photo index, sample index, and README. Use a CC-BY license for eligible research outputs and retain restrictions required for privacy, safety, cultural resources, or land-management compliance.

WORKING DRAFT